

Product Requirements Document (PRD)

1. Product Name

AfyaFlow AI

2. Product Vision

AfyaFlow AI aims to improve patient support and reduce repetitive workload for clinical officers by providing an AI-powered healthcare assistant that offers routine healthcare guidance, appointment support, follow-up reminders, and patient interaction services.

The platform seeks to make healthcare support more accessible, efficient, and responsive, especially in clinics with limited staff resources.

3. Problem Statement

Many healthcare facilities experience high patient volumes and repetitive routine inquiries that consume significant clinical staff time. Patients also face delays in receiving basic healthcare guidance, appointment scheduling, and follow-up support.

There is a need for a simple, accessible, and scalable digital assistant that can support patient interaction while allowing clinical officers to focus on more critical medical tasks.

4. Target Users

Primary Users

- Patients seeking routine healthcare information
- Patients booking appointments
- Patients requiring reminders and follow-up support

Secondary Users

- Clinical officers
 - Small clinics and healthcare facilities
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5. User Pain Points

Patients

- Difficulty accessing quick healthcare guidance
- Missed appointments
- Long waiting times for simple inquiries
- Lack of follow-up reminders

Clinical Officers

- Repetitive patient questions
 - High workload
 - Time spent on non-critical administrative communication
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6. MVP Scope

The Minimum Viable Product (MVP) will include:

- AI-powered chatbot interface
 - Routine healthcare FAQ support
 - Appointment booking functionality
 - Appointment reminder system
 - Responsive web interface
 - English language support
 - Basic patient interaction logging
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7. Out of Scope for Version 1

The following features will NOT be included in the first version:

- Disease diagnosis
- Video consultation
- Payment integration
- Electronic medical records (EMR)

- Voice assistant
 - WhatsApp integration
 - Swahili language support
 - Doctor dashboard
 - Emergency response services
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8. Functional Requirements

Chatbot

- Users can send messages to the chatbot
- Chatbot provides AI-generated healthcare guidance
- Chatbot avoids medical diagnosis
- Chatbot recommends professional medical attention when necessary

Appointment System

- Users can request appointments
- Appointment details are stored in the database
- Users receive reminder notifications

Authentication

- Basic user session management

Database

- Store appointments
 - Store patient interactions
 - Store reminder schedules
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9. Non-Functional Requirements

Performance

- Fast chatbot response time

- Responsive interface on mobile and desktop

Security

- Secure handling of patient interaction data
- API key protection
- Basic authentication security

Usability

- Simple and intuitive interface
- Easy navigation for non-technical users

Scalability

- Backend architecture should support future feature expansion
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10. User Journeys

Patient Asking a Health Question

1. User opens chatbot
 2. User types healthcare question
 3. Chatbot processes request
 4. AI generates response
 5. User receives guidance
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Patient Booking an Appointment

1. User selects appointment option
 2. User enters appointment details
 3. System stores appointment
 4. User receives confirmation
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Appointment Reminder Flow

1. Appointment date approaches
 2. System sends reminder notification
 3. User receives reminder
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11. Success Metrics

The project will be considered successful if:

- Users can successfully interact with the chatbot
 - Appointment booking functions correctly
 - Reminder system works reliably
 - Application is deployed successfully
 - Positive user feedback is received
 - The system reduces repetitive patient inquiries
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12. Risks and Assumptions

Risks

- AI may generate inaccurate healthcare guidance
- Limited access to healthcare datasets
- API costs may increase with scale
- Users may misunderstand AI responses as medical diagnosis

Assumptions

- Users have internet access
 - Clinics are open to digital healthcare support tools
 - OpenAI API remains accessible and reliable
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13. Future Enhancements

Potential future improvements include:

- Swahili language support
- WhatsApp chatbot integration
- Voice assistant
- Clinical officer dashboard
- AI symptom triage system
- Medication reminders
- Analytics dashboard
- Multi-clinic support
- Mobile application
- Integration with healthcare systems